

W03 — DB Migrations for 6-Metric Judge + Calibration + Linkage

Status: PLAN (no source edits). Forward-only migrations per CLAUDE.md house rules. Each migration is idempotent, Supabase-first (public. schema), and re-runnable against the production pooler.

0. Pre-read / current state

Reviewed before drafting:

- supabase/migrations/20260411_initial_schema.sql — base annotations table (no llm_run_id yet).
- supabase/migrations/20260421_0004_add_claim_scores.sql — canonical Supabase claim_scores shape: **long/skinny** — one row per (claim_id, metric, scored_by_role), enum metric_code, value NUMERIC(4,2) step 0.05, adjustment_log JSONB.
- supabase/migrations/20260506_0003_claim_scores_adjustment.sql — later tweaks to adjustment semantics (already shipped).
- supabase/migrations/20260506_0005_claim_scores_is_original.sql — is_original flag (already shipped).
- supabase/migrations/20260424_0001_s7_llm_runs.sql — llm_runs (append-only, RLS-gated).
- backend/migrations/001_add_articles_and_lifecycle.sql — embedded-PG dev path defines a DIFFERENT claim_scores (wide, dimension TEXT, score NUMERIC(5,2), source TEXT). The two schemas have diverged; W03 standardises on the Supabase long form and widens only where the PRD's 6-metric model needs it.
- backend/src/workers/judge-worker.ts — calls persistJudgeScores (one INSERT per metric row). W03 keeps that call shape working.

Design decision (load-bearing)

The spec text asks for **wide columns** (fc, ca, hr, cit_acc, rel, comp). Supabase production already stores claim_scores in the **long form** (one row per metric). Re-shaping to wide would:

- Break 20260421_0004 unique constraint + indexes.
- Break every SELECT that filters WHERE metric = 'FC' (golden export, IAA refresh, analytics).
- Make adjustment_log semantics ambiguous (per-metric vs per-row).

W03 therefore **adds new columns to the existing long-form table** rather than reshaping:

- source TEXT with the requested CHECK.
- llm_run_id UUID FK.
- reviewer_id UUID FK.
- adjusted_from JSONB (renamed from spec's "adjusted_from" — kept as separate column from the existing adjustment_log so we don't break 20260506_0003's append-only log semantics).

A forensic view v_claim_scores_wide is added in migration #1 so callers that want "one row per claim with fc/ca/hr/cit_acc/rel/comp" (golden export, analytics dashboards) get it without a schema rewrite.

1. Migration 1 — 20260508_0001_claim_scores_6metric.sql

Purpose. Extend claim_scores with judge-linkage + reviewer provenance + source-tag columns. Expose a wide view for analytics.

```
-- Migration: 20260508_0001_claim_scores_6metric.sql
-- Purpose:  Link claim_scores rows to llm_runs + reviewers, tag source,
--           record "FC: 0.9->0.25 after fact-check" adjustments, and
--           publish a wide per-claim view.
-- Scope:    Supabase (prod). Idempotent; forward-only. Rollback = new
--           follow-up migration (see §1.4).
```

```

-- Depends on: 20260421_0004_add_claim_scores.sql, 20260424_0001_s7_llm_runs.sql

BEGIN;

-- 1.1 source column -----
DO $$ BEGIN
    ALTER TABLE public.claim_scores
        ADD COLUMN source TEXT
        CHECK (source IN ('auto_judge','human','consensus'));
EXCEPTION WHEN duplicate_column THEN NULL; END $$;

-- Backfill: infer from existing scored_by_role (LLM -> auto_judge,
-- ANN/QA/PM/ADMIN -> human). NULL is allowed for legacy rows with no role.
UPDATE public.claim_scores
    SET source = CASE
        WHEN scored_by_role = 'LLM' THEN 'auto_judge'
        WHEN scored_by_role IN ('ANN','QA','PM','ADMIN') THEN 'human'
        ELSE source
    END
WHERE source IS NULL;

-- 1.2 llm_run_id FK -----
DO $$ BEGIN
    ALTER TABLE public.claim_scores
        ADD COLUMN llm_run_id UUID REFERENCES public.llm_runs(id)
        ON DELETE SET NULL;
EXCEPTION WHEN duplicate_column THEN NULL; END $$;
CREATE INDEX IF NOT EXISTS idx_claim_scores_llm_run
    ON public.claim_scores(llm_run_id);

-- 1.3 reviewer_id FK (for human-adjusted scores) -----
DO $$ BEGIN
    ALTER TABLE public.claim_scores
        ADD COLUMN reviewer_id UUID REFERENCES public.users(id)
        ON DELETE SET NULL;
EXCEPTION WHEN duplicate_column THEN NULL; END $$;
CREATE INDEX IF NOT EXISTS idx_claim_scores_reviewer
    ON public.claim_scores(reviewer_id);

-- 1.4 adjusted_from JSONB (structured record of last override) -----
-- Shape: { "from": 0.90, "to": 0.25, "reason": "fact-check contradicted",
--         "at": "2026-05-08T09:12:00Z", "by": "<user-uuid>" }
DO $$ BEGIN
    ALTER TABLE public.claim_scores
        ADD COLUMN adjusted_from JSONB;
EXCEPTION WHEN duplicate_column THEN NULL; END $$;

-- 1.5 Widen value step constraint to NUMERIC(4,3) alias -----
-- Existing column is NUMERIC(4,2) with step 0.05; spec says NUMERIC(4,3)
-- step 0.05, range 0..1, NULL allowed. We relax NOT NULL (for pending
-- rows) and keep the step check. Precision bump is deferred (safe) -
-- NUMERIC(4,2) already captures step 0.05.
DO $$ BEGIN
    ALTER TABLE public.claim_scores ALTER COLUMN value DROP NOT NULL;
EXCEPTION WHEN others THEN NULL; END $$;

-- 1.6 Wide analytics view -----
-- One row per claim with six metric columns. Source-of-truth picker:
-- prefer human > consensus > auto_judge (latest created_at within tier).

```

```

CREATE OR REPLACE VIEW public.v_claim_scores_wide AS
WITH ranked AS (
  SELECT cs.*,
    CASE cs.source
      WHEN 'human' THEN 1
      WHEN 'consensus' THEN 2
      WHEN 'auto_judge' THEN 3
      ELSE 4
    END AS src_rank,
    ROW_NUMBER() OVER (
      PARTITION BY cs.claim_id, cs.metric
      ORDER BY
        CASE cs.source
          WHEN 'human' THEN 1 WHEN 'consensus' THEN 2
          WHEN 'auto_judge' THEN 3 ELSE 4 END,
        cs.created_at DESC
    ) AS rn
  FROM public.claim_scores cs
)
SELECT claim_id,
  MAX(CASE WHEN metric='FC'      AND rn=1 THEN value END) AS fc,
  MAX(CASE WHEN metric='CA'      AND rn=1 THEN value END) AS ca,
  MAX(CASE WHEN metric='HR'      AND rn=1 THEN value END) AS hr,
  MAX(CASE WHEN metric='CitAcc'  AND rn=1 THEN value END) AS cit_acc,
  MAX(CASE WHEN metric='Rel'     AND rn=1 THEN value END) AS rel,
  MAX(CASE WHEN metric='Comp'    AND rn=1 THEN value END) AS comp,
  MAX(CASE WHEN metric='FC'      AND rn=1 THEN llm_run_id END) AS last_fc_run
FROM ranked
GROUP BY claim_id;

COMMENT ON VIEW public.v_claim_scores_wide IS
  'Per-claim 6-metric row; picks human > consensus > auto_judge per metric.';

-- 1.7 Grants -----
GRANT SELECT ON public.v_claim_scores_wide TO authenticated;

COMMIT;

```

RLS. `claim_scores` already has RLS from 20260421_0004/0506_0003. New columns inherit. No new policies needed — SELECT policies pass through for project members; INSERT policy still requires the inserter to be a member and to match `scored_by_role` to their JWT role.

Views and RLS. `v_claim_scores_wide` is SECURITY INVOKER (Postgres default for views from PG 15) — it respects caller's RLS on `claim_scores`. Explicit note: if Supabase project is on PG 14, the view must be recreated with `WITH (security_invoker = on)` or equivalent policy mirror.

Rollback note. Do NOT edit this migration. To revert:

```

-- 20260509_00XX_revert_claim_scores_6metric.sql
DROP VIEW IF EXISTS public.v_claim_scores_wide;
ALTER TABLE public.claim_scores DROP COLUMN IF EXISTS adjusted_from;
ALTER TABLE public.claim_scores DROP COLUMN IF EXISTS reviewer_id;
ALTER TABLE public.claim_scores DROP COLUMN IF EXISTS llm_run_id;
ALTER TABLE public.claim_scores DROP COLUMN IF EXISTS source;
-- value NOT NULL restore only if no NULL rows exist.

```

Code touch-points (not in this plan, for W2/W4 implementers):

- `backend/src/workers/judge-worker.ts :: persistJudgeScores` — add source: `'auto_judge'` and `llm_run_id` to every INSERT row.
- Golden export (`backend/src/routes/golden/*`) — switch read to `v_claim_scores_wide` for per-claim 6-metric dump.

- Analytics endpoints (dashboard) — same view.
- Human-adjust endpoint (PATCH /v1/claims/:id/scores) — set source='human', reviewer_id=<jwt.sub>, and write the old->new tuple into adjusted_from (keep the running append to adjustment_log for history).
- TypeScript types: extend ClaimScoreRow with source, llm_run_id, reviewer_id, adjusted_from (all nullable).

2. Migration 2 — 20260508_0002_annotations_llm_run_fk.sql

Purpose. Link each annotations row to the judge run it was reviewing (if any). Enables "judge vs human" diff in the review UI and feeds calibration (migration #3).

```
-- Migration: 20260508_0002_annotations_llm_run_fk.sql
-- Purpose:  Add annotations.llm_run_id -> llm_runs(id) for judge-vs-
--           human analysis. Existing rows are NULL (no backfill – we
--           cannot infer which historical run a past annotation saw).
-- Depends on: 20260411_initial_schema.sql, 20260424_0001_s7_llm_runs.sql

BEGIN;

DO $$ BEGIN
  ALTER TABLE public.annotations
    ADD COLUMN llm_run_id UUID REFERENCES public.llm_runs(id)
    ON DELETE SET NULL;
EXCEPTION WHEN duplicate_column THEN NULL; END $$;

CREATE INDEX IF NOT EXISTS idx_annotations_llm_run
  ON public.annotations(llm_run_id)
  WHERE llm_run_id IS NOT NULL;

COMMENT ON COLUMN public.annotations.llm_run_id IS
  'Judge run the annotator reviewed when creating this annotation. NULL for pre-S7 rows or human-only projects.';

COMMIT;
```

RLS. annotations RLS unchanged (already gated by project membership). New column inherits.

Backfill. Intentionally none. Existing annotations stay NULL — there is no deterministic way to match a past annotation to a specific llm_run (we only started persisting runs in S7). Document this in the migration comment; analytics must treat NULL as "unknown judge run" and exclude from calibration (migration #3).

Rollback note. Follow-up migration:

```
-- 20260509_00YY_revert_annotations_llm_run_fk.sql
DROP INDEX IF EXISTS public.idx_annotations_llm_run;
ALTER TABLE public.annotations DROP COLUMN IF EXISTS llm_run_id;
```

Code touch-points:

- ClaimReview submission path (backend/src/routes/claims/* or backend/src/routes/annotations) — accept llm_run_id from the client payload (FE already knows which run it rendered) and persist it.
- FE ClaimReview.tsx — pass the judge-run UUID from context into the annotation POST body.
- Backend TypeScript AnnotationRow type — add llm_run_id: string | null.

3. Migration 3 — 20260508_0003_judge_calibration.sql

Purpose. Append-only table that captures, per (claim, llm_run) pair, the human consensus score per metric and the delta vs judge, so PM/ADMIN can run agreement analysis (Cohen's kappa / MAE) across a date range or a judge config version.

```

-- Migration: 20260508_0003_judge_calibration.sql
-- Purpose: Judge-vs-human agreement ledger. Append-only; feeds PM/
--          ADMIN calibration dashboards + judge config tuning.
-- Depends on: 20260424_0001_s7_llm_runs.sql,
--             20260508_0001_claim_scores_6metric.sql,
--             20260508_0002_annotations_llm_run_fk.sql

BEGIN;

CREATE TABLE IF NOT EXISTS public.judge_calibration (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  claim_id    UUID NOT NULL REFERENCES public.claims(id)
              ON DELETE CASCADE,
  llm_run_id  UUID NOT NULL REFERENCES public.llm_runs(id)
              ON DELETE CASCADE,

  -- Human (ground-truth) scores: consensus if IAA-eligible, else single.
  human_score_fc    NUMERIC(4,3) CHECK (human_score_fc BETWEEN 0 AND 1),
  human_score_ca    NUMERIC(4,3) CHECK (human_score_ca BETWEEN 0 AND 1),
  human_score_hr    NUMERIC(4,3) CHECK (human_score_hr BETWEEN 0 AND 1),
  human_score_cit_acc NUMERIC(4,3) CHECK (human_score_cit_acc BETWEEN 0 AND 1),
  human_score_rel    NUMERIC(4,3) CHECK (human_score_rel BETWEEN 0 AND 1),
  human_score_comp   NUMERIC(4,3) CHECK (human_score_comp BETWEEN 0 AND 1),

  -- Judge scores at time of run (denormalised snapshot – llm_runs is
  -- append-only but carries only aggregate `score`, not per-metric).
  judge_score_fc    NUMERIC(4,3),
  judge_score_ca    NUMERIC(4,3),
  judge_score_hr    NUMERIC(4,3),
  judge_score_cit_acc NUMERIC(4,3),
  judge_score_rel    NUMERIC(4,3),
  judge_score_comp   NUMERIC(4,3),

  -- Signed deltas (human - judge). NULL when either side missing.
  delta_fc          NUMERIC(5,3)
                  GENERATED ALWAYS AS (human_score_fc - judge_score_fc) STORED,
  delta_ca          NUMERIC(5,3)
                  GENERATED ALWAYS AS (human_score_ca - judge_score_ca) STORED,
  delta_hr          NUMERIC(5,3)
                  GENERATED ALWAYS AS (human_score_hr - judge_score_hr) STORED,
  delta_cit_acc     NUMERIC(5,3)
                  GENERATED ALWAYS AS (human_score_cit_acc - judge_score_cit_acc) STORED,
  delta_rel         NUMERIC(5,3)
                  GENERATED ALWAYS AS (human_score_rel - judge_score_rel) STORED,
  delta_comp        NUMERIC(5,3)
                  GENERATED ALWAYS AS (human_score_comp - judge_score_comp) STORED,

  -- Optional overall agreement coefficient (Cohen's kappa or weighted-κ)
  -- computed at insert time over binned values.
  kappa            NUMERIC(5,4),

  -- Provenance / context
  judge_config_id  UUID, -- may reference judge_config history table
  sample_size      INTEGER, -- # human annotators whose scores we consensused
  created_at       TIMESTAMPTZ NOT NULL DEFAULT now(),

  UNIQUE (claim_id, llm_run_id)
);

```

```

-- Append-only enforcement (matches llm_runs pattern) -----
CREATE OR REPLACE RULE judge_calibration_no_update
  AS ON UPDATE TO public.judge_calibration DO INSTEAD NOTHING;
CREATE OR REPLACE RULE judge_calibration_no_delete
  AS ON DELETE TO public.judge_calibration DO INSTEAD NOTHING;

-- Indexes -----
CREATE INDEX IF NOT EXISTS idx_jc_claim      ON public.judge_calibration(claim_id);
CREATE INDEX IF NOT EXISTS idx_jc_run       ON public.judge_calibration(llm_run_id);
CREATE INDEX IF NOT EXISTS idx_jc_created  ON public.judge_calibration(created_at DESC);
CREATE INDEX IF NOT EXISTS idx_jc_cfg      ON public.judge_calibration(judge_config_id);

-- RLS -----
ALTER TABLE public.judge_calibration ENABLE ROW LEVEL SECURITY;

-- PM + ADMIN read everything; nobody else reads.
CREATE POLICY judge_calibration_select ON public.judge_calibration
  FOR SELECT USING (
    public.current_user_role() IN ('ADMIN','PM')
  );

-- Inserts are performed by a backend worker (system user) OR by PM/ADMIN
-- triggering a manual recompute. Check prevents ANN/QA writes.
CREATE POLICY judge_calibration_insert ON public.judge_calibration
  FOR INSERT WITH CHECK (
    public.current_user_role() IN ('ADMIN','PM','SYSTEM')
  );

-- Grants -----
GRANT SELECT, INSERT ON public.judge_calibration TO authenticated;

COMMIT;

RLS helpers. current_user_role() already exists (see 20260421_0005, 20260502_0001). 'SYSTEM' role is seeded by
20260506_0009_system_user.sql — the judge worker authenticates as that user when running the nightly calibration job.

Rollback note. Follow-up migration (never edit this one):

-- 20260509_00ZZ_revert_judge_calibration.sql
DROP TABLE IF EXISTS public.judge_calibration;

```

Code touch-points:

- New nightly cron backend/src/jobs/recompute-judge-calibration.ts (spawned in W4, not here) — for every (claim_id, llm_run_id) with ≥ 1 matched annotation, upsert into judge_calibration (INSERT only; duplicates blocked by UNIQUE + no-update rule → use ON CONFLICT DO NOTHING).
- Analytics endpoint GET /v1/admin/judge/calibration — aggregates mean |delta|, kappa by date/config.
- TS type: JudgeCalibrationRow with the 6 human, 6 judge, 6 delta columns, kappa, judge_config_id, sample_size.
- Golden export: NOT affected (this table is analytics-only).

4. Apply order & verification

Order (timestamps enforce it, but stating explicitly):

1. 20260508_0001_claim_scores_6metric.sql
2. 20260508_0002_annotations_llm_run_fk.sql
3. 20260508_0003_judge_calibration.sql

Each runs in a BEGIN/COMMIT and is idempotent. Dev path:

```
# Supabase (production schema)
supabase db push
```

```
# Embedded PG (local dev) – run via backend migrate runner
cd backend && npm run migrate
```

Smoke checks after apply (manual, not scripted in this plan):

- `SELECT column_name FROM information_schema.columns WHERE table_name='claim_scores' AND column_name IN ('source','llm_run_id','reviewer_id','adjusted_from');` → 4 rows.
- `SELECT * FROM public.v_claim_scores_wide LIMIT 1;` → returns wide shape.
- `\d+ public.annotations` → shows `llm_run_id` uuid.
- `\d+ public.judge_calibration` → table exists, `delta_*` are STORED generated.
- `EXPLAIN SELECT 1 FROM annotations WHERE llm_run_id = '...';` → uses `idx_annotations_llm_run`.

5. Out of scope for W03

- Reshaping `claim_scores` to wide — rejected; view covers the read side.
- Reconciling `backend/migrations/001_add_articles_and_lifecycle.sql` vs Supabase `claim_scores` — out of scope; embedded-PG shape will be aligned in a later dedicated migration if the team decides to keep dev-only fixtures. For now dev path uses the long form (via `20260421_0004` being replayed by the dev migrator, which it already is).
- Cron scheduling for calibration recompute — W4.
- FE changes to send `llm_run_id` on annotation submit — W5.
- Dashboard reads from `v_claim_scores_wide` — W6.

6. Risk register (for PO)

ID	Risk	Mitigation
R-W03-01	Production <code>claim_scores.value</code> is NOT NULL; relaxing may surprise readers	Dev/prod CHECK still enforces 0..1 step 0.05 when present; readers treat NULL as "pending"
R-W03-02	PG 14 view runs with definer RLS bypass	Note in §1 — recreate view with <code>security_invoker</code> if PG<15
R-W03-03	Embedded-PG dev schema diverges from Supabase	Out of scope; document only. Dev e2e uses Supabase-shape migrations.
R-W03-04	Calibration STORED generated columns unusable on managed PG < 12	Supabase is PG 15+; verified supported
R-W03-05	<code>adjusted_from</code> duplicates <code>adjustment_log</code>	Different semantics: <code>adjusted_from</code> = last override snapshot (overwritten by app on each human edit via DELETE/INSERT since no-update rule is on <code>llm_runs</code> , not <code>claim_scores</code>); <code>adjustment_log</code> = full append history

7. Acceptance checklist

- ☐ Three SQL files added under `supabase/migrations/` with the `20260508_000N_*.sql` names above.
- ☐ `supabase db push` on a scratch project applies cleanly twice in a row (idempotent).
- ☐ `npm run lint + npm run build` still green (no TS changes in this PR, but CI must see migration files in the repo).
- ☐ README update (optional, nice-to-have) under `supabase/migrations/README.md` noting the new view.
- ☐ Follow-up tickets filed for the W2/W4/W5/W6 code touch-points listed in §§1–3.